MIL-HDBK-400
NOTICE 1
20 March 1988

MILITARY HANDBOOK

CRITERIA FOR PREPARATION OF GEAR AND SPLINE ENGINEERING DRAWINGS

TO ALL HOLDERS OF MIL-HDBK-400:

1. THE FOLLOWING PAGES OF MIL-HDBK-400 HAVE BEEN REVISED AND SUPERSEDE THE PAGES LISTED:

NEW PAGE	DAGE	SUPERSEDED PAGE	DATE
vii	15 January 1985	vii	Reprinted without change
viii	20 March 1988	viii	15 January 1985
xiii	20 March 1988	viii	15 January 1985
1	15 January 1985	1	Reprinted without change
203	20 March 1988	203	15 January 1985
204	15 January 1985	204	Reprinted without change

2. THE FOLLOWING PAGES HAVE BEEN ADDED TO THE HANDBOOK: 95A-95W INCLUSIVE:

3. RETAIN THIS NOTICE AND INSERT BEFORE TABLE OF CONTENTS.

4. Holders of MIL-HDBK-400 will verify that page additions indicated above have been entered. This notice page will be retained as a check sheet. This issurance, together with appended pages, is a separate publication. Each notice is to be retained by stocking points until the Military Handbook is completely revised or cancelled.

Custodian:
Army – AT
Air Force – 99

Preparing activity:
Army – AT

(Project 3020-0119)

Review activity:
Army – ME

AMSC N/A FSC 3020

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:43Z
Check the source to verify that this is the current version before use.

CONTENTS - Continued

			Page
Paragraph	5.11	Minimum drawing data for parallel sided splines	172
	5.11.1	General	172
	5.11.2	External parallel sided spline data (flat root side fit) . .	172
	5.11.3	Internal parallel sided spline data (flat root side fit) . . .	174
	5.11.4	External parallel sided spline data (flat root major diameter fit)	176
	5.11.5	Internal parallel sided spline data (flat root major diameter fit)	178
	5.11.6	External parallel sided spline data (flat root minor diameter fit)	180
	5.11.7	Internal parallel sided spline data (flat root minor diameter fit)	182
	5.12	Additional and special drawing data for parallel sided splines	184
	5.12.1	General	184
	5.12.2	Spline datum characteristics	184
	5.12.3	Chamfers, radii, and fillets	184
	5.12.4	Shaft end splines	184
	5.12.5	Index relationship	184
	5.12.6	Minor diameter fits	184
	5.12.7	Measurements with pins	184
	5.13	Minimum drawing data for straight sided splines	186
	5.13.1	General	186
	5.13.2	External straight sided spline data (flat root side fit) .	186
	5.13.3	Internal straight sided spline data (flat root side fit) . .	188
	5.13.4	External straight sided spline data (flat root major diameter fit)	190
	5.13.5	Internal straight sided spline data (flat root major diameter fit)	192
	5.13.6	External straight sided spline data (fillet root side fit) . . .	194
	5.13.7	Internal straight sided spline data (fillet root side fit) .	196
	5.14	Additional and special drawing data for straight sided splines	198

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:43Z
Check the source to verify that this is the current version before use.

CONTENTS - Continued

			Page
Paragraph	5.14.1	General	198
	5.14.2	Spline datum characterisitics	198
	5.14.3	Chamfers, radii, and fillets	198
	5.14.4	Shaft end splines	198
	5.14.5	Index relationship	198
	5.14.6	Minor diameter fits	198
	5.14.7	Straight sided/involute mating	198
	5.14.8	Measurements with pins	199
	6.	NOTES	203
	6.1	Supersession data	203
	6.2	Subject term (key word) listing	203

FIGURES

Figures	1a.	External involute spur gear (standard center distance)	9
	1b.	Section X-X	10
	2a.	External involute spur gear (non-standard center distance).	12
	2b.	Section X-X	13
	3a.	External involute helical gear data (standard center distance)	15
	3b.	Section X-X	16
	4a.	External involute helical gear data (non-standard center distance)	18
	4b.	Section X-X	19
	5a.	External involute double helical gear data (standard center distance)	22
	5b.	Section X-X	23
	6a.	Internal involute spur gear data (standard center distance)	25
	6b.	Section X-X	26
	7a.	Internal involute helical gear data (standard center distance)	28
	7b.	Section X-X	29
	8a.	Internal involute double helical gear data (standard center distance)	32
	8b.	Section X-X	33
	9a.	Involute spur rack data	35
	9b.	Section X-X	36

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:43Z
Check the source to verify that this is the current version before use.

CONTENTS - Continued

			Page
Figures	135.	Example drawing of single enveloping worm gear	231
	136.	Example drawing of internal involute clutch spline and external parallel sided spline	232
		TABLES	
Tables	I.	Involute teeth	80
	II.	Standard measuring wire sizes	207
	III.	Involute roll angle table for unit base circle	209
		GEAR DESIGN SHEETS	
Number	1.	90° straight bevel gears	90
	2.	Angular straight bevel gears	91
	3.	90° zerol bevel gears	92
	4.	Angular zerol bevel gears	93
	5.	90° spiral bevel gears	94
	6.	Angular spiral bevel gears	95
	SG-1	External spur gear tooth dimensions standard center distance	95A-95C
	SG-2	External spur gear tooth dimensions non-standard center distance	95D-95G
	SG-3	External spur gear tooth dimensions long addendum - short addendum standard center distance	95H-95K
	HG-1	External helical gear tooth dimensions standard center distance	95L-95O
	HG-2	External helical gear tooth dimensions non-standard center distance	95P-95S
	HG-3	External helical gear tooth dimensions long addendum - short addendum standard center distance	95T-95W
	7.	Over two pins measurement. external involute splines .	170
	8.	Between two pins measurement. internal involute splines	171
		APPENDIX	
Paragraph	10.1	General	204
	10.2	Wire size policy	204
	10.3	Examples of drawing presentation	216

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:43Z
Check the source to verify that this is the current version before use.

1. SCOPE

1.1 Purpose. The purpose of this handbook is to standardize the specification of gears and splines on engineering drawings and related documents. In so doing, it will clarify interpretation of these items making them more suitable for competitive procurement by both, the Government and prime contractors responsible for the quality of gear products procured under subcontracts.

1.2 Scope. This handbook covers parallel, intersecting, and nonintersecting axes gears, whose respective pitch surfaces of revolution are cylinders, cones and hyperboloida. It limits its coverage of splines to involute, parallel sided and straight nonparallel sided tooth forms. It does not attempt to cover either helical or tapered splines. Coverage of gear and spline types is, however, broad enough to permit adequate specification of special gear and spline configurations by supplementing or modifying comparable types within the scope of this handbook.

1.3 Application. This handbook is applicable to all non engineering drawings and related documents utilized in the procurement of gears and splines. It is intended as a reference guide to be used in bot,h the development of new engineering drawings and related documents, and in the review and/or revision of those already .in existence. It is also applicable to any contract in which it is specifically referenced.

1.4 General. Because this handbook will be. utilized in both the specification of new gears and splines as well as updating of existing data. flexibility is sometimes required for the latter. This flexibility is provided by permitting certain types of data to be considered as non-mandatory in the updating of existing data. The type of data under discussion is data not absolutely essential to the manufacture and inspection of the product. While this data may be very helpful in the expansion or alteration of product application by the design engineer. it is often difficult to obtain. Original design criteria and calculations required to establish this data are frequently no longer available. Gear and spline design sheets, formulas. and tables were included in this pamphlet primarily for new designs. However. the user will find them convenient for reestablishing design criteria required for updating existing gear and spline specifications.

1.5 Cost reduction techniques. The use of cost reduction techniques in engineering drawing preparation maybe used when these techniques do not impair the reproducibility quality of MIL-M-9868, clarity and design disclosure requirements for the kind(s) and levels of engineering drawings being prepared. Specifically, techniques currently identified as photographic drafting and use of permanently adhering, non-fading, front printed and mounted decals and paste-ons for repetitive features should be used to the maximum extent practicable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:43Z
Check the source to verify that this is the current version before use.

P		GEAR DESIGN SHEET SG-1 EXTERNAL SPUR GEAR TOOTH DIMENSIONS STANDARD CENTER DISTANCE	DATE:
Ø			SHEET 1 OF 3

Pressure Angle *	Ø		inv Ø		cos Ø		
Diametral Pitch *	P		sin Ø		tan Ø		
Gear Backlash at D	B_G		Circular Pitch	$p = \pi \div P$			
Pinion Backlash at D	B_P		Base Pitch	$p_b = \pi \cos Ø \div P$			
Addendum	a	$1 \div P$	Other		$\cos (90^o/N_G)$		
Dedendum	b	$1.25 \div P$	Other		$\cos (90^o/N_P)$		
Clearance	$\bar{c}$	$\bar{c} = b - a$		π (see note 3) =			
AGMA Quality/Class Number			Measuring Wire Diameter *	d_w			
Gear Major Diameter Runout	Q_G		Gear Major Diameter Tolerance	ΔD_{oG}			
Pinion Major Diameter Runout	Q_P		Pinion Major Diameter Tolerance	ΔD_{oP}			

TERMINOLOGY	FORMULA		SYMBOL	GEAR (G)	PINION (P)
Part Number *			-		
Drawing Number *			-	Rev.	Rev.
Number of Teeth *			N		
Pitch Diameter *	$N \div P$		D		
Base Diameter *	$D \cos Ø$		D_b		
Center Distance *	$(N_G + N_P) \div 2P$		C		+ .00
Tooth Thickness At Pitch Diameter * (Round off to 4 Decimal Places)	$\frac{\pi}{2P} - B_{min}$	Max	t		
	$\frac{\pi}{2P} - B_{max}$	Min			
Major Diameter *	$D + 2a$	Max	D_o		
*	$D_{o_{max}} - \Delta D_o$	Min			
Minimum Tooth Tip Chamfer or Radius Height			Δtip_{min}		
Maximum Tooth Tip Chamfer or Radius Height			Δtip_{max}		
$\sec Ø_o$	$D_{o_{max}} \div D_b$		$\sec Ø_o$		
$\text{inv } Ø_o$	Table of Functions		$\text{inv } Ø_o$		
Tooth Thickness at D_o	$D_{o_{max}}\left(\frac{t_{min}}{D} + \text{inv } Ø - \text{inv } Ø_o\right)$		t_o		

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:43Z
Check the source to verify that this is the current version before use.

P		GEAR DESIGN SHEET SG-1 EXTERNAL SPUR GEAR TOOTH DIMENSIONS STANDARD CENTER DISTANCE	SHEET 2 OF 3
Ø			

TERMINOLOGY	FORMULA	SYMBOL	GEAR (G)	PINION (P)
Top Land	$t_o - 2\,\Delta tip_{max}$ (should be greater than .005)			
Profile Major Diameter *	$D_{o_{max}} - 2\,\Delta tip_{min}$ Max	D_M		
Profile Major Diameter *	$D_{o_{min}} - 2\,\Delta tip_{max}$ Min			
$x = 2C_{min} \sin Ø$	$y = \sqrt{D^2_{MG_{max}} - D^2_{bG}}$	$z = \sqrt{D^2_{MP_{max}} - D^2_{bP}}$		
$x_1 = 2C_{max} \sin Ø$	$y_1 = \sqrt{D^2_{MG_{min}} - D^2_{bG}}$	$z_1 = \sqrt{D^2_{MP_{min}} - D^2_{bP}}$		
NOTE: x and x_1 must be greater than y and y_1 and z and z_1 respectively.				
Active Profile Diameter *	$\sqrt{(x - z)^2 + D^2_{bG}} - Q_P$ Gear	APD		
Active Profile Diameter *	$\sqrt{(x - y)^2 + D^2_{bP}} - Q_G$ Pinion			
Hob Displacement for Backlash	$B_{max} \div 2 \tan Ø$	Δe		
Minor Diameter *	$D - 2b$ Max	D_R		
Minor Diameter *	$D - 2b - 2\,\Delta e$ Min			
Form Diameter Max *	$\sqrt{\left(D_{R_{max}} + 2\bar{c}\right)^2 + \left(\frac{D - D_{R_{max}} - 2\bar{c}}{\tan Ø}\right)^2}$ $\left(D - D_{R_{max}} - 2\bar{c}\right.$ must be a positive value$\left.\right)$	D_F		
Contact ratio	$\left(y_1 + z_1 - x_1\right) \div 2p_b$ To assure continuity of action, value must be greater than 1.05, preferably 1.2 or greater.	m_p		
W_G ; W_P (may be positive or negative values)	$W_G = \text{inv } Ø + \frac{d_w}{D_{bG}} - \frac{\pi}{N_G}$	W_G		
	$W_P = \text{inv } Ø + \frac{d_w}{D_{bP}} - \frac{\pi}{N_P}$	W_P		

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:43Z
Check the source to verify that this is the current version before use.

P		GEAR DESIGN SHEET SG-1 EXTERNAL SPUR GEAR TOOTH DIMENSIONS STANDARD CENTER DISTANCE			SHEET 3 OF 3
Ø					

TERMINOLOGY	FORMULA		SYMBOL	GEAR (G)	PINION (P)
Involute at Wire Center	$W_G + \frac{t_{G_{max}}}{D_G}$; $W_P + \frac{t_{P_{min}}}{D_P}$	Max	inv $Ø_w$		
	$W_G + \frac{t_{G_{min}}}{D_G}$; $W_P + \frac{t_{P_{min}}}{D_P}$	Min			
sec $Ø_w$	Table of Functions	Max / Min	sec $Ø_w$		
tan $Ø_{w_{max}}$	Table of Functions		tan $Ø_w$		
Wire Contact Diameter	$\sqrt{(D_b \tan Ø_w - d_w)^2 + D_b^2}$		D_c		
Measurement over * Wires (Even Teeth) *	$D_b \sec Ø_w + d_w$	Max / Min	M_E		
Wire Clearance	$d_{w_c} = .5(M_{E_{min}} - 2d_w - D_{R_{max}})$ (value must be positive)		d_{w_c}		
Measurement over * Wires (Odd Teeth) *	$D_b \cos(90^o/N) \sec Ø_w + d_w$	Max / Min	M_o		
Face Width *			F		

COARSE PITCH GEARS		GEAR (G)	PINION (P)
Maximum Runout Tolerance of Pitch Diameter	*		
Lead Tolerance Across Face Width	*		
Pitch (Tooth-to-Tooth Spacing) Tolerance	*		
Profile Tolerance	*		

NOTES:

1. All values designated with an asterisk (*) shall be shown on drawing.
2. To insure accuracy, calculate to a minimum of seven decimal places.
3. Use π to a minimum of seven decimal places.
4. These calculations were performed to _____ decimal places.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:43Z
Check the source to verify that this is the current version before use.

P		GEAR DESIGN SHEET SG-2 EXTERNAL SPUR GEAR TOOTH DIMENSIONS NON-STANDARD CENTER DISTANCE	DATE:
Ø			SHEET 1 OF 4

Pressure Angle *	Ø		inv Ø		cos Ø		
Diametral Pitch *	P		tan Ø				
Gear Backlash at D_x	B_{xG}		Circular Pitch at D_x	$p_x = \pi D_x \div N$			
Pinion Backlash at D_x	B_{xP}		Base Pitch	$p_b = \pi \cos Ø \div P$			
Addendum	a	$1 \div P$	Other		$\cos (90^o/N_G)$		
Dedendum	b	$1.25 \div P$	Other		$\cos (90^o/N_P)$		
Clearance	$\bar{c}$	$\bar{c} = b - a$		Measuring Wire Diameter *	d_w		
AGMA Quality/Class Number			Operating Center Distance *	C_x	+.00		
Operating Pitch Diameter *	$2NC_x \div (N_G + N_P)$		D_{xG}		D_{xP}		
$\sec Ø_x = D_x \div D_b$			Operating Pressure Angle *	$Ø_x$			
π (see note 3) =			inv $Ø_x$		$\sin Ø_x$		
Gear Major Diameter Runout	Q_G		Gear Major Diameter Tolerance	ΔD_{oG}			
Pinion Major Diameter Runout	Q_P		Pinion Major Diameter Tolerance	ΔD_{oP}			

TERMINOLOGY	FORMULA	SYMBOL	GEAR (G)	PINION (P)
Part Number *		-		
Drawing Number *		-	Rev.	Rev.
Number of Teeth *		N		
Nominal Pitch Diameter *	$N \div P$	D		
Base Diameter *	$D \cos Ø$	D_b		
Actual Tooth Thickness at D_x	$.5p_x - B_{xG_{min}}$; $.5p_x - B_{xP_{min}}$ Max $.5p_x - B_{xG_{max}}$; $.5p_x - B_{xP_{max}}$ Min	t_x		
Actual Tooth Thickness at D * (Round off to 4 Decimal Places) *	$D\left(\frac{t_x}{D_x} + \text{inv } Ø_x - \text{inv } Ø\right)$ Max / Min	t		
Hob Displacement for Backlash	$\left(t - \frac{\pi}{2P}\right) \div 2 \tan Ø$ Max / Min	Δe		

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:43Z
Check the source to verify that this is the current version before use.

P		GEAR DESIGN SHEET SG-2 EXTERNAL SPUR GEAR TOOTH DIMENSIONS NON-STANDARD CENTER DISTANCE			SHEET 2 OF 4
Ø					

TERMINOLOGY	FORMULA		SYMBOL	GEAR (G)	PINION (P)
Major Diameter *	$D + 2a + 2\Delta e_{max}$	Max	D_o		
*	$D_{o_{max}} - \Delta D_o$	Min			
Minimum Tooth Tip Chamfer or Radius Height			Δtip_{min}		
Maximum Tooth Tip Chamfer or Radius Height			Δtip_{max}		
$\sec \emptyset_o$	$D_{o_{max}} \div D_b$		$\sec \emptyset_o$		
$\text{inv } \emptyset_o$	Table of Functions		$\text{inv } \emptyset_o$		
Tooth Thickness at D_o	$D_{o_{max}}\left(\frac{t_{min}}{D} + \text{inv } \emptyset - \text{inv } \emptyset_o\right)$		t_o		
Top Land	$t_o - 2\Delta tip_{max}$ (should be greater than .005)		L_T		
Profile Major Diameter *	$D_{o_{max}} - 2\Delta tip_{min}$	Max	D_M		
*	$D_{o_{min}} - 2\Delta tip_{max}$	Min			
$x = 2C_{x_{min}} \sin \emptyset_x$	$y = \sqrt{D^2_{MG_{max}} - D^2_{bG}}$		$z = \sqrt{D^2_{MP_{max}} - D^2_{bP}}$		
$x_1 = 2C_{x_{max}} \sin \emptyset_x$	$y_1 = \sqrt{D^2_{MG_{min}} - D^2_{bG}}$		$z_1 = \sqrt{D^2_{MP_{min}} - D^2_{bP}}$		
NOTE: x and x_1 must be greater than y and y_1 and z and z_1 respectively.					
Active Profile Diameter *	$\sqrt{(x - z)^2 + D^2_{bG}} - Q_P$	Gear	APD		
*	$\sqrt{(x - y)^2 + D^2_{bP}} - Q_G$	Pinion			
Nominal Tooth Thickness at D	$D\left(\frac{.5p_x}{D_x} + \text{inv } \emptyset_x - \text{inv } \emptyset\right)$		t_{nom}		
Hob Displacement at Zero Backlash	$\left(t_{nom} - \frac{\pi}{2P}\right) \div 2 \tan \emptyset$		Δe_{nom}		
Minor Diameter *	$D - 2b + 2\Delta e_{nom}$	Max	D_R		
*	$D - 2b + 2\Delta e_{min}$	Min			

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:43Z
Check the source to verify that this is the current version before use.

P		GEAR DESIGN SHEET SG-2 EXTERNAL SPUR GEAR TOOTH DIMENSIONS NON-STANDARD CENTER DISTANCE		SHEET 3 OF 4	
Ø					
TERMINOLOGY		FORMULA	SYMBOL	GEAR (G)	PINION (P)
Form Diameter Maximum *		$\sqrt{\left(D_{R_{max}} + 2\bar{c}\right)^2 + \left(\frac{D - D_{R_{max}} - 2\bar{c}}{\tan \phi}\right)^2}$ $\left(D - D_{R_{max}} - 2\bar{c}\right.$ must be a positive value$\left.\right)$	D_F		
Contact Ratio		$\left(y_1 + z_1 - x_1\right) \div 2p_b$ To assure continuity of action, value must be greater than 1.05 preferably 1.2 or greater	m_p		
W_G ; W_P (may be a positive or negative value)		$W_G = \text{inv } \phi + \frac{d_w}{D_{bG}} - \frac{\pi}{N_G}$ $W_P = \text{inv } \phi + \frac{d_w}{D_{bP}} - \frac{\pi}{N_P}$	W_G W_P		
Involute at Wire Center		$W_G + \frac{t_{G_{max}}}{D_G}$; $W_P + \frac{t_{P_{max}}}{D_P}$ $W_G + \frac{t_{G_{min}}}{D_G}$; $W_P + \frac{t_{P_{min}}}{D_P}$	Max / Min inv ϕ_w		
sec ϕ_w		Table of Functions	Max / Min sec ϕ_w		
tan $\phi_{w_{max}}$		Table of Functions	tan ϕ_w		
Wire Contact Diameter		$\sqrt{\left(D_b \tan \phi_w - d_w\right)^2 + D_b^2}$	D_c		
Measurement Over Wires (Even Teeth) *		$D_b \sec \phi_w + d_w$	Max / Min M_E		
Wire Clearance		$d_{w_c} = .5\left(M_{e_{min}} - 2d_w - D_{R_{max}}\right)$ (value must be positive)	d_{w_c}		
Measurement over Wires (Odd Teeth) *		$D_b \cos(90^o/N) \sec \phi_w + d_w$	Max / Min M_o		
Face Width *			F		

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:43Z
Check the source to verify that this is the current version before use.

P		GEAR DESIGN SHEET SG-2 EXTERNAL SPUR GEAR TOOTH DIMENSIONS NON-STANDARD CENTER DISTANCE	SHEET 4 OF 4
Ø			

COARSE PITCH GEARS		GEAR (G)	PINION (P)
Maximum Runout Tolerance of Pitch Diameter	*		
Lead Tolerance Across Face Width	*		
Pitch (Tooth-to-Tooth Spacing) Tolerance	*		
Profile Tolerance	*		

NOTES:

1. All values designated with an asterisk (*) shall be shown on drawing.
2. To insure accuracy, calculate to a minimum of seven decimal places.
3. Use π to a minimum of seven decimal places.
4. These calculations were performed to _____ decimal places.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:43Z
Check the source to verify that this is the current version before use.

P		GEAR DESIGN SHEET SG-3 EXTERNAL SPUR GEAR TOOTH DIMENSIONS LONG ADDENDUM–SHORT ADDENDUM STANDARD CENTER DISTANCE	DATE:
Ø			SHEET 1 OF 4

Pressure Angle *	Ø		inv Ø		cos Ø		
Diametral Pitch *	P		sin Ø		tan Ø		
Gear Backlash at D	B_G		Circular Pitch	$p = \pi \div P$			
Pinion Backlash at D	B_P		Base Pitch	$p_b = \pi \cos Ø \div P$			
Addendum	a	$1 \div P$	Other		$\cos (90^o/N_G)$		
Dedendum	b	$1.25 \div P$	Other		$\cos (90^o/N_P)$		
Clearance	$\bar{c}$	$\bar{c} = b - a$		π (see note 3) =			
AGMA Quality/Class Number			Measuring Wire Diameter *	d_w			
Gear Major Diameter Runout	Q_G		Gear Major Diameter Tolerance	ΔD_{oG}			
Pinion Major Diameter Runout	Q_P		Pinion Major Diameter Tolerance	ΔD_{oP}			

TERMINOLOGY	FORMULA	SYMBOL	GEAR (G)	PINION (P)
Part Number *		-		
Drawing Number *		-	Rev.	Rev.
Number of Teeth *		N		
Pitch Diameter *	$N \div P$	D		
Base Diameter *	$D \cos Ø$	D_b		
Center Distance *	$(N_G + N_P \div 2P)$	C		+.00
Hob Displacement (Show Plus or Minus)	$e_G = -e_P$; $e_{P_{max}} = .5D_G \sin^2 Ø - a$ $e_P \leqq e_{P_{max}}$; $e_{P_{min}} = a - .5D_P \sin^2 Ø$ $e_{P_{min}} \angle e_P \angle e_{P_{max}}$	e		
Hob Displacement for Backlash	$B_{min} \div 2 \tan Ø$ Min $B_{max} \div 2 \tan Ø$ Max	Δe		
Total Hob Displacement (may be plus or minus)	$e - \Delta e_{min}$ Max $e - \Delta e_{max}$ Min	E		

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:43Z
Check the source to verify that this is the current version before use.

P		GEAR DESIGN SHEET SG-3 EXTERNAL SPUR GEAR TOOTH DIMENSIONS LONG ADDENDUM-SHORT ADDENDUM STANDARD CENTER DISTANCE			SHEET 2 OF 4
Ø					

TERMINOLOGY	FORMULA		SYMBOL	GEAR (G)	PINION (P)
Tooth Thickness at Pitch Diameter (Round off to 4 Decimal Places) *	$.5p + 2E_{max} \tan Ø$	Max	t		
*	$.5p + 2E_{min} \tan Ø$	Min			
Major Diameter *	$D + 2a + 2e$	Max	D_o		
*	$D_{o_{max}} - \Delta D_o$	Min			
Minimum Tooth Tip Chamfer or Radius Height			Δtip_{min}		
Maximum Tooth Tip Chamfer or Radius Height			Δtip_{max}		
$\sec Ø_o$	$D_{o_{max}} \div D_b$		$\sec Ø_o$		
$\text{inv } Ø_o$	Table of Functions		$\text{inv } Ø_o$		
Tooth Thickness at D_o	$D_{o_{max}} \left(\frac{t_{min}}{D} + \text{inv } Ø - \text{inv } Ø_o \right)$		t_o		
Top Land	$t_o - 2\Delta tip_{max}$ (should be greater than .005)		L_T		
Profile Major Diameter *	$D_{o_{max}} - 2\Delta tip_{min}$	Max	D_M		
*	$D_{o_{min}} - 2\Delta tip_{max}$	Min			

$x = 2C_{min} \sin Ø$		$y = \sqrt{D^2_{MG_{max}} - D^2_{bG}}$		$z = \sqrt{D^2_{MP_{max}} - D^2_{bP}}$	
$x_1 = 2C_{max} \sin Ø$		$y_1 = \sqrt{D^2_{MG_{min}} - D^2_{bG}}$		$z_1 = \sqrt{D^2_{MP_{min}} - D^2_{bP}}$	

NOTE: x and x_1 must be greater than y and y_1 and z and z_1 respectively.

Active Profile Diameter *	$\sqrt{(x - z)^2 + D^2_{bG}} - Q_P$	Gear	APD		
*	$\sqrt{(x - y)^2 + D^2_{bP}} - Q_G$	Pinion			

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:43Z
Check the source to verify that this is the current version before use.

P		GEAR DESIGN SHEET SG-3 EXTERNAL SPUR GEAR TOOTH DIMENSIONS LONG ADDENDUM-SHORT ADDENDUM STANDARD CENTER DISTANCE SHEET 3 OF 4			
Ø					
TERMINOLOGY		FORMULA		SYMBOL	GEAR (G)	PINION (P)
Minor Diameter	*	$D - 2b + 2e$	Max	D_R		
	*	$D - 2b + 2E_{min}$	Min			
Form Diameter Maximum	*	$\sqrt{\left(D_{R_{max}} + 2\bar{c}\right)^2 + \left(\frac{D - D_{R_{max}} - 2\bar{c}}{\tan \emptyset}\right)^2}$ $\left(D - D_{R_{max}} - 2\bar{c}\right.$ must be a positive value$\left.\right)$		D_F		
Contact Ratio		$\left(y_1 + z_1 - x_1 \div 2p_b\right)$ To assure continuity of action, value must be greater than 1.05 preferably 1.2 or greater		m_p		
W_G ; W_P (may be a positive or negative value)		$W_G = \text{inv } \emptyset + \frac{d_w}{D_{bG}} - \frac{\pi}{N_G}$ $W_P = \text{inv } \emptyset + \frac{d_w}{D_{bP}} - \frac{\pi}{Np}$		W_G W_P		
Involute at Wire Center		$W_G + \frac{t_{G_{max}}}{D_G}$; $W_P + \frac{t_{P_{max}}}{D_P}$	Max	$\text{inv } \emptyset_w$		
		$W_G + \frac{t_{G_{min}}}{D_G}$; $W_P + \frac{t_{P_{min}}}{Dp}$	Min			
$\sec \emptyset_w$		Table of Functions	Max Min	$\sec \emptyset_w$		
$\tan \emptyset_{w_{max}}$		Table of Functions		$\tan \emptyset_w$		
Wire Contact Diameter		$\sqrt{\left(D_b \tan \emptyset_w - d_w\right)^2 + D_b^2}$		D_c		
Measurement Over Wires (Even Teeth)	*	$D_b \sec \emptyset_w + d_w$	Max	M_E		
	*		Min			
Wire Clearance		$d_{w_c} = .5\left(M_{e_{min}} - 2d_w - D_{R_{max}}\right)$ (value must be positive)		d_{w_c}		

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:43Z
Check the source to verify that this is the current version before use.

P		GEAR DESIGN SHEET SG-3 EXTERNAL SPUR GEAR TOOTH DIMENSIONS LONG ADDENDUM-SHORT ADDENDUM STANDARD CENTER DISTANCE	SHEET 4 OF 4
Ø			

TERMINOLOGY	FORMULA	SYMBOL	GEAR (G)	PINION (P)
Measurement over * Wires (Odd * Teeth)	$D_b \cos(90^o/N)\sec\emptyset_w + d_w$ Max / Min	M_o		
Face Width *		F		

COARSE PITCH GEARS	GEAR (G)	PINION (P)
Maximum Runout Tolerance of Pitch Diameter *		
Lead Tolerance Across Face Width *		
Pitch (Tooth-to-Tooth Spacing) Tolerance *		
Profile Tolerance *		

NOTES:

1. All values designated with an asterisk (*) shall be shown on drawing.
2. To insure accuracy, calculate to a minimum of seven decimal places.
3. Use π to a minimum of seven decimal places.
4. These calculations were performed to _____ decimal places.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:43Z
Check the source to verify that this is the current version before use.

P_n / ϕ_n		GEAR DESIGN SHEET HG-1 EXTERNAL HELICAL GEAR TOOTH DIMENSIONS STANDARD CENTER DISTANCE	DATE: SHEET 1 OF 4

Normal Diametral Pitch *	P_n		Helix Angle *	ψ			
Normal Pressure Angle *	ϕ_n		$\sin\psi$		$\cos\psi$		
$\tan\phi = \tan\phi_n \div \cos\psi$			$\sin\phi_n$		$\tan\phi_n$		
Transverse Pressure Angle	ϕ		$\cos\psi_b = \sin\phi_n \div \sin\phi$				
Normal Backlash of Gear	B_{nG}		$\sin\phi$		$\cos\phi$		
Normal Backlash of Pinion	B_{nP}		$\text{inv}\,\phi$		π (see note 3) =		
Addendum	a	$1 \div P_n$	Other		Normal Circular Pitch	$p_n = \pi \div P_n$	
Dedendum	b	$1.25 \div P_n$	Other		Circular Pitch	$p = \pi \div P_n \cos\psi$	
Clearance	$\overline{c}$	$\overline{c} = b - a$			AGMA Quality/Class Number		
Base Pitch	$p_b = p\cos\phi$				Measuring Wire Diameter *	d_w	
Gear Major Diameter Runout	Q_G				Gear Major Diameter Tolerance	ΔD_{oG}	
Pinion Major Diameter Runout	Q_P				Pinion Major Diameter Tolerance	ΔD_{oP}	

TERMINOLOGY	FORMULA	SYMBOL	GEAR (G)	PINION (P)
Part Number *		-		
Drawing Number *		-	Rev.	Rev.
Number of Teeth *		N		
Pitch Diameter *	$N \div P_n \cos\psi$	D		
Base Diameter *	$D\cos\phi$	D_b		
Center Distance *	$(N_G + N_P) \div 2P_n \cos\psi$	C		+.00
Normal Tooth Thickness at Pitch Diameter (Round off to 4 Decimal Places) *	$.5p_n - B_{nG_{min}}$; $.5p_n - B_{nP_{min}}$ Max $.5p_n - B_{nG_{max}}$; $.5p_n - B_{nP_{max}}$ Min	t_n		
Transverse Arc Tooth Thickness	$t_{nG_{max}} \div \cos\psi$; $t_{nP_{max}} \div \cos\psi$ Max $t_{nG_{min}} \div \cos\psi$; $t_{nP_{min}} \div \cos\psi$ Min	t		

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:43Z
Check the source to verify that this is the current version before use.

P_n		GEAR DESIGN SHEET HG-1 EXTERNAL HELICAL GEAR TOOTH DIMENSIONS STANDARD CENTER DISTANCE			SHEET 2 OF 4
$\emptyset_n$					

TERMINOLOGY	FORMULA	SYMBOL	GEAR (G)	PINION (P)
Major Diameter *	$D + 2a$ Max	D_o		
*	$D_{o_{max}} - \Delta D_o$ Min			
Minimum Tooth Tip Chamfer or Radius Height		Δtip_{min}		
Maximum Tooth Tip Chamfer or Radius Height		Δtip_{max}		
sec $\emptyset_o$	$D_{o_{max}} \div D_b$	sec $\emptyset_o$		
inv $\emptyset_o$	Table of Functions	inv $\emptyset_o$		
Tooth Thickness at D_o	$D_{o_{max}} \left(\frac{t_{min}}{D} + \text{inv } \emptyset - \text{inv } \emptyset_o \right)$	t_o		
Top Land	$t_o - 2\Delta\, tip_{max}$ (should be greater than .010)	L_T		
Profile Major Diameter *	$D_{o_{max}} - 2\Delta\, tip_{min}$ Max	D_M		
*	$D_{o_{min}} - 2\Delta\, tip_{max}$ Min			

$x = 2C_{min} \sin \emptyset$		$y = \sqrt{D^2_{MG_{max}} - D^2_{bG}}$		$z = \sqrt{D^2_{MP_{max}} - D^2_{bP}}$	
$x_1 = 2C_{max} \sin \emptyset$		$y_1 = \sqrt{D^2_{MG_{min}} - D^2_{bG}}$		$z_1 = \sqrt{D^2_{MP_{min}} - D^2_{bP}}$	

NOTE: x and x_1 must be greater than y and y_1 and z and z_1 respectively.

Active Profile Diameter *	$\sqrt{(x - z)^2 + D^2_{bP}} - Q_P$ Gear	APD		
*	$\sqrt{(x - y)^2 + D^2_{bP}} - Q_G$ Pinion			
Hob Displacement for Backlash	$B_{nG_{max}} \div 2\tan \emptyset_n$; $B_{nP_{max}} \div 2\tan \emptyset_n$	Δe		
Minor Diameter *	$D - 2b$ Max	D_R		
*	$D - 2b - 2\Delta e$ Min			

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:43Z
Check the source to verify that this is the current version before use.

P_n		GEAR DESIGN SHEET HG-1 EXTERNAL HELICAL GEAR TOOTH DIMENSIONS STANDARD CENTER DISTANCE			SHEET 3 OF 4
ϕ_n					
TERMINOLOGY		FORMULA		SYMBOL	GEAR (G)	PINION (P)
Form Diameter * Max		$\sqrt{\left(D_{R_{max}} + 2\bar{c}\right)^2 + \left(\frac{D - D_{R_{max}} - 2\bar{c}}{\tan \phi}\right)^2}$ ($D - D_{R_{max}} - 2\bar{c}$ must be a positive value)		D_F		
Face Width *				F		
Total Contact Ratio		$\left[\left(y_1 + z_1 - x_1\right) \div 2p_b\right] + \frac{F \sin \psi}{P_n}$		m_p		
W_G ; W_P (may be positive or negative values)		$W_G = \text{inv } \phi + \frac{d_w \sec \psi_b}{D_{bG}} - \frac{\pi}{N_G}$		W_G		
		$W_P = \text{inv } \phi + \frac{d_w \sec \psi_b}{D_{bP}} - \frac{\pi}{N_P}$		W_P		
Involute at Wire Center		$W_G + \frac{t_{G_{max}}}{D_G}$; $W_P + \frac{t_{P_{max}}}{D_P}$	Max	$\text{inv } \phi_w$		
		$W_G + \frac{t_{G_{min}}}{D_G}$; $W_P + \frac{t_{P_{min}}}{D_P}$	Min			
$\sec \phi_w$		Table of Functions	Max / Min	$\sec \phi_w$		
$\tan \phi_{w_{max}}$		Table of Functions		$\tan \phi_w$		
Wire Contact Diameter		$\sqrt{\left(D_b \tan \phi_w - d_w \cos \psi_b\right)^2 + D_b^2}$		D_c		
Measurement * Over Wires * (Even Teeth)		$D_b \sec \phi_w + D_w$	Max / Min	M_E		
Wire Clearance		$d_{w_c} = .5\left(M_{E_{min}} - 2d_w - D_{R_{max}}\right)$ (value must be positive)		d_{w_c}		

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:43Z
Check the source to verify that this is the current version before use.

P_n		GEAR DESIGN SHEET HG-1 EXTERNAL HELICAL GEAR TOOTH DIMENSIONS STANDARD CENTER DISTANCE	SHEET 4 OF 4
$\emptyset_n$			

TERMINOLOGY	FORMULA	SYMBOL	GEAR (G)	PINION (P)
Measurement * Over One Wire * (Odd Teeth)	$\frac{1}{2} M_E$ Max / Min	M_o		
Lead *	$\pi N \div P_n \sin \psi$	L		

COARSE PITCH GEARS		GEAR (G)	PINION (P)
Maximum Runout Tolerance of Pitch Diameter	*		
Lead Tolerance Across Face Width	*		
Pitch (Tooth-To-Tooth Spacing) Tolerance	*		
Profile Tolerance	*		

NOTES:

1. All values designated with an asterisk (*) shall be shown on drawing.
2. To insure accuracy, calculate to a minimum of seven decimal places.
3. Use π to a minimum of seven decimal places.
4. These calculations were performed to _____ decimal places.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:43Z
Check the source to verify that this is the current version before use.

P_n		GEAR DESIGN SHEET HG-2 EXTERNAL HELICAL GEAR TOOTH DIMENSIONS NON-STANDARD CENTER DISTANCE	DATE:
ϕ_n			SHEET 1 OF 4

Normal Diametral Pitch *	P_n		Helix Angle *	ψ			
Normal Pressure Angle *	ϕ_n		$\sin \psi$		$\cos \psi$		
$\sin \phi_n$		$\tan \phi_n$		$\tan \phi = \tan \phi_n \div \cos \psi$			
$\cos \phi_n$		Transverse Pressure Angle	ϕ		$\cos \phi$		
inv ϕ		$\sin \phi$		$\cos \psi_b = \sin \phi_n \div \sin \phi$			
Normal Gear Backlash at D	B_{nG}		Addendum	a	$1 \div P_n$	Other	
Normal Pinion Backlash at D	B_{nP}		Dedendum	b	$1.25 \div P_n$	Other	
Operating Center Distance *	C_x	+.00	Clearance	$\bar{c}$	b - a		
Operating Pitch Diameter *	$2NC_x \div (N_G + N_P)$		D_{xG}		D_{xP}		
Circular Pitch	$p = \pi \div P_n \cos \psi$		Base Pitch	$p_b = p \cos \phi$			
Normal Circular Pitch	$p_n = \pi \div P_n$		π (see note 3) =				
Circular Pitch at D_x	$p_x = \pi D_x \div N$		$\cos \phi_x = D_b \div D_x$				
Operating Pressure Angle *	ϕ_x		$\sin \phi_x$		$\text{inv } \phi_x$		
AGMA Quality/Class Number			Measuring Wire Diameter *	d_w			
Gear Major Diameter Runout	Q_G		Gear Major Diameter Tolerance	ΔD_{oG}			
Pinion Major Diameter Runout	Q_P		Pinion Major Diameter Tolerance	ΔD_{oP}			

TERMINOLOGY	FORMULA	SYMBOL	GEAR (G)	PINION (P)
Part Number *		-		
Drawing Number *		-	Rev.	Rev.
Number of Teeth *		N		
Pitch Diameter *	$N \div P_n \cos \psi$	D		
Base Diameter *	D cos ϕ	D_b		
Backlash In Transverse Plane at D_x	$B_{x_{min}} = \frac{B_{n_{min}} \cos \phi}{\cos \phi_n \cos \psi \cos \phi_x}$ (Min) $B_{x_{max}} = \frac{B_{n_{max}} \cos \phi}{\cos \phi_n \cos \psi \cos \phi_x}$ (Max)	B_x		

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:43Z
Check the source to verify that this is the current version before use.

P_n		GEAR DESIGN SHEET HG-2 EXTERNAL HELICAL GEAR TOOTH DIMENSIONS NON-STANDARD CENTER DISTANCE			SHEET 2 OF 4	
ϕ_n						
TERMINOLOGY		**FORMULA**		**SYMBOL**	**GEAR (P)**	**PINION (P)**
Actual Transverse Tooth Thickness at D_x		$.5p_x - B_{x_{min}}$ $.5p_x - B_{x_{max}}$	Max Min	t_x		
Normal Tooth Thickness at D (Round off to 4 Decimal Places)	* *	$D\left(\frac{t_x}{D_x} + \text{inv}\,\phi_x - \text{inv}\,\phi\right)\cos\psi$	Max Min	t_n		
Total Hob Displacement		$\left(t_n - \frac{\pi}{2P_n}\right) \div 2\tan\phi_n$	Max Min	Δe		
Major Diameter	* *	$D + 2a + 2\Delta e_{max}$ $D_{o_{max}} - \Delta D_o$	Max Min	D_o		
Minimum Tooth Tip Chamfer or Radius Height				Δtip_{min}		
Maximum Tooth Tip Chamfer or Radius Height				Δtip_{max}		
sec ϕ_o		$D_{o_{max}} \div D_b$		sec ϕ_o		
inv ϕ_o		Table of Functions		inv ϕ_o		
Tooth Thickness at D_o		$D_{o_{max}}\left(\frac{t_{n_{min}}}{D\cos\psi} + \text{inv}\,\phi - \text{inv}\,\phi_o\right)$		t_o		
Top Land		$t_o - 2\,\Delta tip_{max}$ (should be greater than .010)		L_T		
Profile Major Diameter	* *	$D_{o_{max}} - 2\,\Delta tip_{min}$ $D_{o_{min}} - 2\,\Delta tip_{max}$	Max Min	D_M		
$x = 2C_{x_{min}} \sin\phi_x$		$y = \sqrt{D^2_{MG_{max}} - D^2_{bG}}$		$z = \sqrt{D^2_{MP_{max}} - D^2_{bP}}$		
$x_1 = 2C_{x_{max}} \sin\phi_x$		$y_1 = \sqrt{D^2_{MG_{min}} - D^2_{bG}}$		$z_1 = \sqrt{D^2_{MP_{min}} - D^2_{bP}}$		
NOTE: x and x_1 must be greater than y and y_1 and z and z_1 respectively.						

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:43Z
Check the source to verify that this is the current version before use.

P_n		GEAR DESIGN SHEET HG-2 EXTERNAL HELICAL GEAR TOOTH DIMENSIONS NON-STANDARD CENTER DISTANCE			SHEET 3 OF 4
ϕ_n					

TERMINOLOGY	FORMULA	SYMBOL	GEAR (G)	PINION (P)
Active Profile Diameter *	$\sqrt{(x - z)^2 + D_{bG}^2} - Q_P$ Gear	APD		
*	$\sqrt{(x - y)^2 + D_{bP}^2} - Q_G$ Pinion			
Normal Nominal Tooth Thickness at D	$D\left(\frac{.5p_x}{D_x} + \text{inv}\,\phi_x - \text{inv}\,\phi\right)\cos\psi$	t_{n_N}		
Hob Displacement at Zero Backlash	$\left(t_{n_N} - \frac{\pi}{2P_n}\right) \div 2\tan\phi_n$	Δe_{nom}		
Minor Diameter *	$D - 2b + 2\,\Delta e_{nom}$ Max	D_R		
*	$D - 2b + 2\,\Delta e_{min}$ Min			
Form Diameter Max *	$\sqrt{\left(D_{R_{max}} + 2\bar{c}\right)^2 + \left(\frac{D - D_{R_{max}} - 2\bar{c}}{\tan\phi}\right)^2}$ ($D - D_{R_{max}} - 2\bar{c}$ must be a positive value)	D_F		
Face Width *		F		
Total Contact Ratio	$\left[\left(y_1 + z_1 - x_1\right) \div 2p_b\right] + \frac{F \sin\psi}{p_n}$	m_p		
W_G ; W_P (may be positive or negative values)	$W_G = \text{inv}\,\phi_x + \frac{d_w}{D_{bG}\cos\psi_b} - \frac{\pi}{N_G}$	W_G		
	$W_P = \text{inv}\,\phi_x + \frac{d_w}{D_{bP}\cos\psi_b} - \frac{\pi}{N_P}$	W_P		
Involute at Wire Center	$W_G + \frac{t_{xG_{max}}}{D_{xG}}$; $W_P + \frac{t_{xP_{max}}}{D_{xP}}$ Max	$\text{inv}\,Q_w$		
	$W_G + \frac{t_{xG_{min}}}{D_{xG}}$; $W_P + \frac{t_{xP_{min}}}{D_{xP}}$ Min			

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:43Z
Check the source to verify that this is the current version before use.

P_n		GEAR DESIGN SHEET HG-2 EXTERNAL HELICAL GEAR TOOTH DIMENSIONS NON-STANDARD CENTER DISTANCE	SHEET 4 OF 4
ϕ_n			

TERMINOLOGY	FORMULA	SYMBOL	GEAR (G)	PINION (P)
sec ϕ_w	Table of Functions Max / Min	sec O_w		
tan $\phi_{w_{max}}$	Table of Functions	tan ϕ_w		
Wire Contact Diameter	$\sqrt{(D_b \tan \phi_w - d_w \cos \psi_b)^2 + D_b^2}$	D_c		
Measurement Over * Wires (Even Teeth) *	$D_b \sec \phi_w + d_w$ Max / Min	M_E		
Wire Clearance	$d_{w_c} = (.5\ M_{E_{min}} - 2d_w - D_{R_{max}})$ (value must be positive)	d_{w_c}		
Measurement Over * One Wire (Odd Teeth) *	$\frac{1}{2} M_E$ Max / Min	M_o		
Lead *	$\pi N \div P_n \sin \psi$	L		

COARSE PITCH GEARS		GEAR (G)	PINION (P)
Maximum Runout Tolerance of Pitch Diameter	*		
Lead Tolerance Across Face Width	*		
Pitch (Tooth-to-Tooth Spacing) Tolerance	*		
Profile Tolerance	*		

NOTES:

1. All values designated with an asterisk (*) shall be shown on drawing.
2. To insure accuracy, calculate to a minimum of seven decimal places.
3. Use π to a minimum of seven decimal places.
4. These calculations were performed to _____ decimal places.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:43Z
Check the source to verify that this is the current version before use.

P_n		GEAR DESIGN SHEET HG-3 EXTERNAL HELICAL GEAR TOOTH DIMENSIONS LONG ADDENDUM-SHORT ADDENDUM STANDARD CENTER DISTANCE	DATE:
ϕ_n			SHEET 1 OF 4

Normal Diametral Pitch *	P_n		Helix Angle *	ψ			
Normal Pressure Angle *	ϕ_n		$\sin\psi$		$\cos\psi$		
$\tan\phi = \tan\phi_n \div \cos\psi$			$\sin\phi_n$		$\tan\phi_n$		
Transverse Pressure Angle	ϕ		$\cos\psi_b = \sin\phi_n \div \sin\phi$				
Normal Gear Backlash at D	B_{nG}		$\sin\phi$		$\cos\phi$		
Normal Pinion Backlash at D	B_{nP}		$\text{inv}\,\phi$		π (see note 3) =		

Addendum	a	$1 \div P_n$	Other		Normal Circular Pitch	$p_n = \pi \div P_n$		
Dedendum	b	$1.25 \div P_n$	Other		Circular Pitch	$p = \pi \div P_n \cos\psi$		
Clearance	$\overline{c}$	$\overline{c} = b - a$			Measuring Wire Diameter *	d_w		
Base Pitch	$p_b = p \cos\phi$				AGMA Quality/Class Number			

Gear Major Diameter Runout	Q_G		Gear Major Diameter Tolerance	ΔD_{oG}	
Pinion Major Diameter Runout	Q_P		Pinion Major Diameter Tolerance	ΔD_{oP}	

TERMINOLOGY	FORMULA	SYMBOL	GEAR (G)	PINION (P)
Part Number *		-		
Drawing Number *		-	Rev.	Rev.
Number of Teeth *		N		
Pitch Diameter *	$N \div P_n \cos\psi$	D		
Base Diameter *	$D \cos\phi$	D_b		
Center Distance *	$(N_G + N_P) \div 2P_n \cos\psi$	C	+.00	
Hob Displacement (Show plus or minus)	$e_G = -e_P$; $e_{P_{max}} = .5D_G\sin^2\phi_n - a$ $e_P \leqq e_{P_{max}}$; $e_{P_{min}} = a - .5D_P\sin^2\phi_n$	e		
Hob Displacement for Backlash	$B_{nG_{min}} \div 2\tan\phi_n ; B_{nP_{min}} \div 2\tan\phi_n$ Min $B_{nG_{max}} \div 2\tan\phi_n ; B_{nP_{min}} \div 2\tan\phi_n$ Max	Δe		

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:43Z
Check the source to verify that this is the current version before use.

P_n		GEAR DESIGN SHEET HG-3 EXTERNAL HELICAL GEAR TOOTH DIMENSIONS LONG ADDENDUM-SHORT ADDENDUM STANDARD CENTER DISTANCE			SHEET 2 OF 4
ϕ_n					

TERMINOLOGY	FORMULA		SYMBOL	GEAR (G)	PINION (P)
Total Hob Displacement (may be plus or minus)	$e - \Delta e_{min}$ $e - \Delta e_{max}$	Max Min	E		
Normal Tooth * Thickness at D * (Round off to 4 Decimal Places)	$.5p_n + 2E_{max} \tan \phi_n$ $.5p_n + 2E_{min} \tan \phi_n$	Max Min	t_n		
Transverse Arc Tooth Thickness	$t_{n_{max}} \div \cos \psi$ $t_{n_{min}} \div \cos \psi$	Max Min	t		
* Major Diameter *	$D + 2a + 2e$ $D_{o_{max}} - \Delta D_o$	Max Min	D_o		
Minimum Tooth Tip Chamfer or Radius Height			Δtip_{min}		
Maximum Tooth Tip Chamfer or Radius Height			Δtip_{max}		
sec ϕ_o	$D_{o_{max}} \div D_b$		sec ϕ_o		
inv ϕ_o	Table of Functions		inv ϕ_o		
Tooth Thickness at D_o	$D_{o_{max}} \left(\frac{t_{min}}{D} + \text{inv } \phi - \text{inv } \phi_o \right)$		t_o		
Top Land	$t_o - 2\Delta tip_{max}$ (should be greater than .010)		L_T		
* Profile Major Diameter *	$D_{o_{max}} - 2\Delta tip_{min}$ $D_{o_{min}} - 2\Delta tip_{max}$	Max Min	D_M		

$x = 2C_{min} \sin \phi$		$y = \sqrt{D^2_{MG_{max}} - D^2_{bG}}$		$z = \sqrt{D^2_{MP_{max}} - D^2_{bP}}$	
$x_1 = 2C_{max} \sin \phi$		$y_1 = \sqrt{D^2_{MG_{min}} - D^2_{bG}}$		$z_1 = \sqrt{D^2_{MP_{min}} - D^2_{bP}}$	

NOTE: x and x_1 must be greater than y and y_1 and z and z_1 respectively.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:43Z
Check the source to verify that this is the current version before use.

P_n		GEAR DESIGN SHEET HG-3 EXTERNAL HELICAL GEAR TOOTH DIMENSIONS LONG ADDENDUM-SHORT ADDENDUM STANDARD CENTER DISTANCE			SHEET 3 OF 4
ϕ_n					

TERMINOLOGY	FORMULA		SYMBOL	GEAR (G)	PINION (P)
Active Profile Diameter *	$\sqrt{(x - z)^2 + D_{bG}^2} - Q_P$	Gear	APD		
*	$\sqrt{(x - y)^2 + D_{bP}^2} - Q_G$	Pinion			
Minor Diameter *	$D - 2b + 2e$	Max	D_R		
*	$D - 2b + 2E_{min}$	Min			
Form Diameter Max *	$\sqrt{\left(D_{R_{max}} + 2\bar{c}\right)^2 + \left(\frac{D - D_{R_{max}} - 2\bar{c}}{\tan \phi}\right)^2}$ ($D - D_{R_{max}} - 2\bar{c}$ must be a positive value)		D_F		
Face Width *			F		
Total Contact Ratio	$\left[\left(y_1 + z_1 - x_1\right) \div 2p_b\right] + \frac{F \sin \psi}{p_n}$		m_p		
W_G ; W_P (may be positive or negative values)	$W_G = \text{inv } \phi + \frac{d_w}{D_{b_G} \cos \psi_b} - \frac{\pi}{N_G}$		W_G		
	$W_P = \text{inv } \phi + \frac{d_w}{D_{b_P} \cos \psi_b} - \frac{\pi}{N_P}$		W_P		
Involute at Wire Center	$W_G + \frac{t_{G_{max}}}{D_G}$; $W_P + \frac{t_{P_{max}}}{D_P}$	Max	$\text{inv } \phi_w$		
	$W_G + \frac{t_{G_{min}}}{D_G}$; $W_P + \frac{t_{P_{min}}}{D_P}$	Min			
sec ϕ_w	Table of Functions	Max Min	sec ϕ_w		
tan $\phi_{w_{max}}$	Table of Functions		tan ϕ_w		

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:43Z
Check the source to verify that this is the current version before use.

P_n		GEAR DESIGN SHEET HG-3 EXTERNAL HELICAL GEAR TOOTH DIMENSIONS LONG ADDENDUM-SHORT ADDENDUM STANDARD CENTER DISTANCE	SHEET 4 OF 4
ϕ_n			

TERMINOLOGY	FORMULA	SYMBOL	GEAR (G)	PINION (P)
Wire Contact Diameter	$\sqrt{\left(D_b \tan \phi_w - d_w \cos \psi_b\right)^2 + D_b^2}$	D_c		
Measurement Over Wires (Even Teeth) *	$D_b \sec \phi_w + d_w$ Max / Min	M_E		
Wire Clearance	$d_{w_c} = .5\left(M_{E_{min}} - 2d_w - D_{R_{max}}\right)$ (value must be positive)	d_{w_c}		
Measurement Over One Wire (Odd Teeth) *	$\frac{1}{2} M_E$ Max / Min	M_o		
Lead *	$\pi N \div P_n \sin \psi$	L		

COARSE PITCH GEARS		GEAR (G)	PINION (P)
Maximum Runout Tolerance of Pitch Diameter	*		
Lead Tolerance Across Face Width	*		
Pitch (Tooth-to-Tooth Spacing) Tolerance	*		
Profile Tolerance	*		

NOTES:

1. All values designated with an asterisk (*) shall be shown on drawing.
2. To insure accuracy, calculate to a minimum of seven decimal places.
3. Use π to a minimum of seven decimal places.
4. These calculations were performed to _____ decimal places.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:43Z
Check the source to verify that this is the current version before use.

6. NOTES

6.1 Supersession data. This handbook supersedes USA TACOM Pamphlet No. 11-46, 31 August 1973.

6.2 Subject term (key word) listing.

Engineering drawings, gear and spline
Gear and spline, preparation of engineering drawings
Spline and gear, preparation of engineering drawings

Custodian:
Army - AT
Air Force - 99

Review activities:
Army - ME

Preparing activity:
Army - AT

(Project 3020-0109)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:43Z
Check the source to verify that this is the current version before use.

APPENDIX

10.1 General. For convenience, the appendix contains tables, examples, formulas, etc., frequently used by the designer in the preparation of gear and spline drawings.

a. It is recommended that designers familiarize themselves with the instructions outlined in the text prior to the use of the tables and formulas.

b. To assist the designer in the interpretation of the requirements of this handbook, several examples of drawings are included in the appendix. These sample drawings shall be construed as informational only and are complete to the degree necessary to illustrate a condition. Actual drawings shall comply with the instructions set forth in the handbook.

c. Specification of measuring wire or ball sizes on drawings shall be to sizes listed in the table and/or according to the instructions contained in the appendix.

d. Involute profile controls require the specification of roll angles, as in profile charts. A table of roll angles is listed here for the convenience of the user.

e. Specific instructions for the use of all tables and formulas are outlined in the text.

10.2 Wire size policy. Recent practice has shown that measurement of gears, splines, and serrations is frequently accomplished through the use of two or three identical sized wires or pins. This method necessitates the use of proper diameter of measuring wire or pin in order to obtain reliable readings. The reasons for using this method are cost and procurement considerations, especially for small quantities. It is sometimes used in the early production of large orders.

a. It is desirable to use wires that contact the profile at or near the pitch diameters and/or reference diameter. With such wires the measurement at or near the pitch diameter is least affected by profile deviations. To illustrate this, consider figure 116 which shows a full line depicting a true outline and a dotted line showing a profile of the same pitch diameter but with a profile deviation. The same conditions would affect circular gears, splines, and serrations. The wire (B) contacts both outlines at or near the pitch diameter, and the measurement will be affected very little by profile deviations. It is evident from the same figure that a maximum wire (M) will give false results unless the profile deviation is considered in the computation.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:43Z
Check the source to verify that this is the current version before use.